

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1102

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing Techniques (BL4,BL5)
Assessment Tool Weightage: 35%

QUESTIONS: 5

Total Marks:35

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

*I M Nithishkumaran 192320037(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Nithishkumaran

RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Constraint-Based Problem Solving

Q4. Scenario: Smart Library Management System

Features:

Book Search, Book Reservation, Borrow/Return, Notifications

Usability Constraints:

- Search results displayed within 2 seconds
- Accessible interface for all age groups
- Minimal learning curve for first-time users

Question:

Conduct a usability evaluation using appropriate testing methods. Analyze how the given constraints impact interface design and recommend improvements to enhance usability and accessibility. **(BL5)**

1. Problem Understanding

The Smart Library Management System consists of Book Search, Book Reservation, Borrow/Return, and Notifications. The system should provide a simple and efficient interface for library users.

The main usability constraints are:

- Search results must be displayed within 2 seconds.
- The interface should be accessible to all age groups.
- The system should have a minimal learning curve for first-time users.

2. Constraint Analysis

Search Performance:

Search results should be displayed within 2 seconds. Slow results can reduce user satisfaction and efficiency.

Accessibility:

The interface should use readable text, clear buttons, simple navigation and understandable messages so that users of different age groups can use it comfortably.

Learning Curve:

First-time users should be able to search, reserve, borrow and return books without requiring extensive training or assistance.

3. Usability Testing Methods

User Testing:

Users are asked to perform common tasks such as searching for a book, checking availability and making a reservation. Their difficulties and errors are observed.

Task-Based Testing:

Users are given specific tasks and their completion time, number of errors and need for assistance are recorded.

Accessibility Testing:

The interface is evaluated for readability, button clarity, navigation and ease of interaction for different age groups.

First-Time User Testing:

New users are allowed to use the system without detailed instructions. Their ability to complete basic tasks is observed.

Response-Time Testing:

Search operations are tested to ensure that results are displayed within the required 2 seconds.

4. Important Test Cases

Test Case	Expected Result
Search for an existing book	Results displayed within 2 seconds
Search for unavailable book	Clear availability status displayed
Reserve an available book	Reservation completed successfully
Borrow/return a book	Book status updated correctly
View notification	Notification is clear and understandable
First-time user searches for a book	User completes task with minimal assistance
User navigates through the system	Functions are easy to locate

5. Usability Challenges

- Slow search results.
- Complicated navigation.
- Small or unclear buttons.
- Difficult-to-read information.
- Confusing reservation and return procedures.
- Unclear error or notification messages.
- Too many steps for simple operations.

6. Recommended Improvements

- Provide a clearly visible search bar on the main screen.
- Display book title, author and availability clearly.
- Use simple options such as Search, Reserve, Borrow/Return and Notifications.
- Use readable fonts and clearly labelled buttons.
- Reduce unnecessary steps in common operations.
- Provide simple confirmation and error messages.
- Optimize the search function to maintain the 2-second response requirement.

7. Justification

The proposed testing methods directly address the given constraints. Response-time testing verifies the 2-second search requirement, while accessibility and user testing determine whether the interface can be comfortably used by different age groups. First-time-user and task-based testing verify whether users can perform basic library operations without extensive training.

Therefore, these techniques help make the Smart Library Management System fast, accessible, simple and user-friendly, while satisfying the specified usability constraints.

8. Conclusion

The Smart Library Management System should provide a fast, accessible and easy-to-learn interface. The given constraints directly determine the testing priorities: search performance must be verified against the 2-second requirement, accessibility must be evaluated with users from different age groups, and first-time-user testing must confirm that the system can be learned with minimal assistance.

By applying these usability testing methods and implementing improvements such as simple navigation, readable interfaces, clear availability information, understandable notifications and optimized search, the system can provide a more efficient, accessible and user-friendly library experience.